

Ancient Script Decipherment
Phase 3 Report
From-Scratch Model Stack, Trained on CLSP

Waiz Khan
JHU CLSP — advisor: Philipp Koehn

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1 Summary

Four model families were implemented from scratch, smoke-verified on CPU, and trained for real on cluster GPUs: a sealed character LM, a BPE encoder–decoder translation control, the pixel encoder–decoder, and masked-window self-supervised pretraining. Every parameter was randomly initialized and trained inside this repository; the no-pretrained CI gate guards the sealed tree. The full-size pixel model is $\sim 97\text{M}$ parameters, inside the 50–400M target band.

2 Trained models

model	family	steps	wall	best held-out CE
Hebrew sealed char-LM	sealed-lm-hebrew	13,000	~ 4 h	1.5109
Coptic sealed char-LM	sealed-lm-coptic	15,000	~ 5 h	1.2582
Greek sealed char-LM	sealed-lm-greek	42,000	21.4 h	1.5994
BPE control (Egy $\rightarrow$ De)	mt-tla-earlier-bpe	12,000	9.7 h	5.5635

Table 1: All runs early-stopped on held-out cross-entropy and preregistered in the ledger. These are *not* headline numbers: one seed each, and no frozen test partition has been read. The three sealed LMs ($\sim 10.7\text{M}$ params each) are the only language models the decipherment track may consume.

The pixel kill-gate (run 1709819, 200 steps) reached held-out CE 6.3034 after rendering TLA hieroglyphs to pixels on a compute node — the first end-to-end GPU execution of the project’s central mechanism. The BPE control’s converged 5.5635 is the line the pixel model must beat, at three seeds, before anything is claimed.

3 Architecture

Pre-LayerNorm Transformer blocks with sinusoidal positions and the Salesky configuration (12 encoder / 3 decoder layers, $d = 512$, FF 4096, 4 heads). The pixel front end embeds each 24×24 window (Conv2D $3 \times 3 \rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ linear projection); the BPE control swaps in a source embedding matrix and shares everything else, so the comparison is architecture-matched by construction. Target subwords come from a SentencePiece *trainer* run on our own corpora. The

training loop is explicit PyTorch: Adam with linear warmup then inverse-sqrt decay, gradient accumulation, clipping, bf16 autocast on CUDA with fp32 fallback, DDP wrapping from the SLURM environment, early stopping, and rank-0 checkpointing.

Self-supervised pretraining uses masked-window reconstruction (MAE family) — chosen as the exact from-scratch analog of the pretrained PIXEL model used by the published LogogramNLP baseline, and because stroke-level detail is the signal for writing. JEPA-style latent prediction and DINO-style self-distillation, implemented from scratch, are queued as Phase 7 ablations; pretrained DINOv2/v3 *checkpoints* remain banned from the sealed pipeline and are permitted only as labeled open-book comparisons (Phase 8).

4 What the kill-gates caught

The mandatory one-budget-unit runs found four real defects, each costing seconds instead of hours:

1. torch 2.13 silently fell back to CPU on older cluster drivers (49 minutes of fake “GPU” training). Linux now pins torch 2.4.1+cu118, which runs on every driver generation in the fleet.
2. GPU jobs now assert `torch.cuda.is_available()` before launching — no silent CPU training, ever.
3. Node c04 throws `CUDA unknown error` on contact and is excluded from all jobs.
4. Paragraph-length Greek asked cairo for a surface wider than it will allocate; `render_strip` now truncates to the window budget before allocating (no information lost — `slice_windows` already caps at `max_windows`). Regression test added.

5 Next: Phase 4

chrF-primary translation metrics, Ithaca-style restoration with damage-shaped masks from the Phase 2 degradation suite, conformal candidate sets with measured coverage, and the auxiliary heads — the first evaluations to touch the frozen test partitions.